

Scalability & Future Plan

Date	2 Jul 2024
Team ID	XXXXXX
Project Name	OptiCrop
Maximum Marks	1 Mark

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	The application is designed for local deployment and does not support multiple concurrent users.	Limits accessibility and scalability.	High
2	The machine learning model is trained on a single dataset, which may limit prediction accuracy for diverse agricultural conditions.	May reduce recommendation accuracy for different regions.	High
3	The system does not integrate real-time weather or soil data.	Predictions are based only on user-provided inputs.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports local users through the Flask development server.	Deploy the application on a cloud platform such as Render or AWS to support multiple users.
2	Data Storage	Uses a CSV dataset and a trained model file (<code>crop_model.pkl</code>).	Integrate a database (MySQL or PostgreSQL) for storing datasets, user history, and prediction records.
3	Performance	Prediction is performed using a locally stored machine learning model.	Optimize the model, improve preprocessing, and use caching to reduce response time.
4	Security	No user authentication or secure communication is implemented.	Implement user authentication, HTTPS, and secure input validation to improve application security.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Deploy the application to a cloud platform and improve the user interface.	Next 2–3 Months	Increased accessibility and improved user experience.
Phase 3	Integrate real-time weather APIs and soil data for more accurate recommendations.	Next 4–6 Months	Improved prediction accuracy and more reliable recommendations.
Phase 4	Add user accounts, prediction history, multilingual support, and advanced analytics dashboards.	Future Enhancement	Better usability, personalized recommendations, and wider adoption.